The food model: a semantic assessment and proposals (R53, M100)

Subject: specification version 3 (`food_model_spec.md`). Symbols and equation numbers are the specification's. Nothing here is implemented: each proposal is for decision, and an accepted proposal enters the specification's change log first.

0. Criteria

Every equation is tested against four questions:

Criterion	Question
(C1) Dimensions	Are the terms of every sum in the same unit, so that adding them means something?
(C2) Meaning	Does the quantity measure what its name says?
(C3) Representation	Which real mechanism that matters for the outcome is missing?
(C4) Identifiability	Can each parameter be measured or sourced, or does it depend on an arbitrary choice?

1. Findings

#	Equation	Finding	Criterion
F1	(15) $K_g^{(1)} = s_g + \nu_g$	adds a satiety ratio (0.9-2.3) to a nutrient adequacy score (0-1); the implicit weight 1 : 1 has no meaning	C1, C4
F2	(13) ν_g	a property of the food alone. A food's nutritional value depends on what the diet already lacks: vitamin C from lemons is worth nothing to a person already covered, everything to one with scurvy	C2, C3
F3	(13) ν_g	all 13 nutrients weigh the same: sodium counts as much as B12 or iodine, whose lack causes far worse harm	C2
F4	(13) ν_g	needs are scaled per kcal with a fixed $E^{\text{ref}} = 2600$, while the person's energy need E^{need} varies from 2,350 to 3,450 kcal. Needs for vitamins and minerals do not scale with energy	C2

#	Equation	Finding	Criterion
F5	(12) s_g and (11) σ_g	satiety appears twice: as preference (s_g, a protein-based fit to Holt) and as fullness (σ_g, Holt's index). Protein also appears a third time in ν_g	C2
F6	(14) β_g	well-being per kcal. A lemon has few kcal, so its well-being per kcal is the highest of all goods; pleasure comes per serving, not per kcal. Fat enters three times, alcohol only as a benefit (its harm at higher doses is absent)	C1, C2, C3
F7	M5	well-being is computed and spent (pass 2) but never measured: there is no outcome for it, so it can feed nothing	C3
F8	Algorithm 1	pass 1 ignores well-being; pass 2 ignores nutrition. A fed person still short of vitamin C takes sugar over cabbage in pass 2. Pass 1 stops at "fed" even when nutrient gaps remain	C2
F9	(17), (30)	diminishing returns are counted in servings m_g while steps are in kcal. By (P2) each good's total value scales with $e_g m_g$ (kcal per serving), so changing the definition of a serving changes what people eat	C4
F10	(7) $P^{\max} = 0.35E^{\max}/4$	the protein ceiling is tied to the energy ceiling, not to what is eaten. The fishing village eats 50 % of its energy as protein without consequence; the source (Bilsborough and Mann) states a share of energy eaten, and an absolute liver capacity	C2
F11	(6) $F^{\max} = \bar{\sigma} E^{\max}$	the fullness capacity rests on the mean of the 38 foods Holt happened to test (1.36). A different food list gives a different capacity; Holt measured two hours after one serving, not a day	C4

#	Equation	Finding	Criterion
F12	(25)-(26)	all energy eaten above the basal rate becomes activity ($\Lambda^{\text{act}} = E/B$); there is no body store. Fat reserves that carry people through lean seasons, and that surplus becomes, are missing	C3
F13	(26) $work$	one factor for all work. Desk work does not scale with surplus energy as hauling does	C2
F14	(4) caffeine	caffeine only raises the energy need. Its effect on effort and alertness, the reason it matters (M85), is not in any outcome	C3
F15	(23) W^{drink}	grams of a drink times the hydration index. The beverage hydration index is retention relative to water over hours, measured by urine output (PMID 42884730), not water content: milk is 88 % water and is counted as 150 %	C1, C2
F16	W^{need} constant	the water need does not rise with work and heat, while the energy need does	C3
F17	(1) a_g	stocks are spread evenly over the year. Fresh goods (fruit, milk in season, fresh fish) come in seasons; vitamin C deficiency is seasonal	C3
F18	the whole model	one representative person per city. Hunger and malnutrition fall on the poor first; a city whose average is fed can have a starving underclass	C3
F19	the whole model	choice depends on availability only. Once the goods economy has prices, the budget decides what the poor eat	C3

2. Proposals

Each proposal states the reasoning, the new equations, and the variables it adds, removes or changes.

Proposal A. Nutrition as diet adequacy, driven by the gaps (fixes F1-F5, F9 for nutrition)

Reasoning:

- What makes a food nutritionally valuable at a given moment is how much it closes what the diet still lacks.
- A static per-food score (13) cannot express that. A diet-level adequacy function can, and its marginal value per good then changes as the diet fills.
- Weighting each nutrient by the harm its lack causes ties nutrition to the deficiency burden the population model already uses.
- Protein enters as digestible indispensable protein (protein × DIAAS), so protein quality lives in one place.
- Per-good bioavailability carries the PubMed findings of `round_m84.md`: iron, zinc, niacin, carotene.

Equations:

$$c_i(\mathbf{q}) = \frac{1}{r_i} \sum_g \frac{q_g \eta_g}{100}\, b_{gi}\, n_{gi}, \qquad A(\mathbf{q}) = \sum_{i \in \mathcal{I}_0} \omega_i \, \min\bigl(1,\, c_i(\mathbf{q})\bigr)$$

- $\mathcal{I}_0 = \mathcal{I} \cup \{\text{energy}\}$, with $c_{\text{energy}} = E / E^{\text{need}}$ and $n_{g,\text{protein}} = \pi_g \min(\delta_g, 1)$.
- The marginal value of good g per kcal is the partial derivative

$$\frac{\partial A}{\partial (e_g q_g)} = \frac{1}{e_g} \sum_{i:\, c_i < 1} \omega_i \, \frac{\eta_g \, b_{gi} \, n_{gi}}{100 \, r_i}.$$

It falls to zero for every nutrient already covered. Diminishing returns then come from the gaps closing, not from servings.

Variables:

Change	Variable	Note
removed	ν_g (13), s_g (12), $K_g^{(1)}$ (15), E^{ref}, $\kappa_s, \bar{p}, \bar{\delta}$ (in the preference)	replaced
removed	m_g in pass 1	servings no longer drive nutritional diminishing returns
added	ω_i	weight of nutrient i: the deficiency-burden weights of the population model (`round_m83.md` section 2c); energy and protein weighted highest (protein leverage, PMID 37820518)
added	b_{gi}	bioavailability of nutrient i in good g: plant iron 0.05-0.12 against mixed 0.14-0.18 (Hurrell and Egli 2010); cereal niacin about 1/3 unless nixtamalized (Carpenter 1983); carotene 1/2 to 1/10; B12 by food (Watanabe 2007)
added	$A(\mathbf{q})$	diet adequacy, 0 to $\sum \omega_i$
changed	c_i	from an outcome (22) to a state updated during the allocation; (21) folds into b_{gi}

What changes in results:

- A city with milk and grain, short of vitamin C, takes its apples and potatoes for vitamin C until it is covered, then stops valuing them for it.
- Beef keeps its value while B12, iron or zinc are short.
- Sodium stops lifting meat once salt covers it.

Proposal B. Well-being per serving, with sensory-specific satiety and an outcome (fixes F6, F7)

Reasoning:

- Pleasure comes in eating events. It falls with repetition of the same food: sensory-specific satiety, a short-term satiety signal (PMID 1752065). So well-being belongs per serving, with diminishing returns per food, not per kcal.
- Alcohol pleases at low doses and harms at higher ones (a biphasic dopamine response, PMID 39729041), so it needs a harm term.
- Fat should enter once, as reward (with carbohydrate) and as heaviness, not three times.
- The result must be an outcome, Ψ, so that it can feed the happiness line.

Equations:

$$\Psi(\mathbf{q}) = \sum_g h_g \ln\!\left(1 + \frac{q_g}{m_g}\right) \;-\; \zeta \,\max\!\left(0,\; \textstyle\sum_g q_g \eta_g \alpha_g / 100 - \alpha^{\text{safe}}\right)$$

$$h_g = u_1 z_g + u_2 f_g k_g + u_3 \left[\alpha_g > 0\right] - u_5 \,\mathrm{ED}_g \qquad (\text{per serving; } \mathrm{ED}_g = \text{energy density, kcal/g})$$

Variables:

Change	Variable	Note
removed	β_g per kcal (14), the lightness term from water per kcal	replaced; lightness becomes low energy density, the property Wells et al. measured as heaviness
added	h_g	hedonic value of one serving of g
added	Ψ	well-being attained per person a day: an outcome, to feed the happiness line (one happy face per step of Ψ, to be set)
added	$\zeta,\ \alpha^{\text{safe}}$	alcohol harm slope and safe daily alcohol (to source)
changed	m_g	now only in Ψ, where a serving has a meaning: one eating event

Proposal C. One consumption program with two priorities (fixes F8, F9; replaces Algorithm 1)

Reasoning:

- Your rule is two separate variables in two sorted orders: nutrition first, then well-being (“steak and potatoes, then dessert”).
- The exact mathematical form of “first A, then B, without giving up A” is a lexicographic optimisation.
- Stage 1 closes energy and nutrient gaps as far as the goods and the stomach allow.
- Stage 2 adds what pleases, in the room left, without undoing stage 1.
- This removes the arbitrary stop of pass 1 at “fed”. People keep eating for nutrition while gaps remain, and well-being never displaces a nutrient they still lack.

Equations:

$$\begin{aligned}
&\text{Stage 1:} \quad A^* = \max_{\mathbf{q}} \; A(\mathbf{q}) \\
&\text{Stage 2:} \quad \max_{\mathbf{q}} \; \Psi(\mathbf{q}) \quad \text{subject to} \quad A(\mathbf{q}) \geq A^*, \\
&\text{both stages:} \quad 0 \leq q_g \leq \bar{q}_g, \quad F(\mathbf{q}) \leq F^{\max}, \quad E(\mathbf{q}) \leq E^{\max}, \quad P \leq P^{\text{abs}} \text{ (Proposal E)}, \quad R \leq Q^{\text{fruit}}.
\end{aligned}$$

Solution method: the same step-by-step choice as now. Stage 1 always takes the good with the largest $\partial A / \partial (e_g q_g)$; stage 2 the good with the largest $\partial \Psi$ among steps that keep A. A is concave (a sum of capped linear terms) and Ψ is concave, so taking the steepest admissible step is the standard method.

Variables:

Change	Variable	Note
removed	passes $\rho \in \{1, 2\}$ with stops F^{fed} and $F^{\max}$; keys $K^{(1)}, K^{(2)}$	replaced by the two stages
added	A^*	the best adequacy the goods allow
changed	fed (19)	stays as an outcome, F / F^{fed}, no longer a stopping rule

Proposal D. One satiety construct (fixes F5, F11)

Reasoning:

- Fullness (how much a person can eat) and nutrition (what they need) are different things. Holt's index measures the first; it should appear once, as fullness.
- The protein factor s_g was a stand-in for Holt (fitted to beef against bread). Under Proposal A, protein's pull is in A through ω_{protein}, so s_g has no role left.
- The fullness capacity should not depend on which foods Holt chose to test. Calibrating it so that a reference diet is exactly fed at its need makes it a property of people, not of a food list.

Equations:

$$F^{\max} = \left(\frac{\sum_g \sigma_g e_g q_g^{\text{ref}}}{\sum_g e_g q_g^{\text{ref}}} \right) E^{\max}$$

Here q^{ref} is a documented reference diet: for example a sourced pre-industrial English or modern mixed diet.

Variables:

Change	Variable	Note
removed	s_g, κ_s, $\bar{p}$, $\bar{\delta}$	not needed under A
changed	$\bar{\sigma}$	from Holt's sample mean (1.36) to the mean fullness of a reference diet q^{ref}
added	q_g^{ref}	a sourced reference diet (data to find)

Proposal E. Protein: an absolute ceiling and a share-based harm (fixes F10)

Reasoning: Bilsborough and Mann (2006) give two separate facts:

- a liver capacity: a theoretical safe maximum of 285-365 g a day for 80 kg, about 3.6-4.6 g per kg;
- a danger when protein exceeds 35 % of energy eaten.

The first is a hard ceiling. The second is a harm that rises with the share, not a reason to stop eating.

Equations:

$$P \le P^{\text{abs}} = 3.6\ \text{g/kg} \times 63.3\ \text{kg} \approx 228\ \text{g}, \qquad \text{burden}_{\text{protein}} = \psi\ \max\!\Big(0,\ \frac{4P}{E} - 0.35\Big)$$

63.3 kg is the EFSA mean adult weight.

Variables:

Change	Variable	Note
changed	$P^{\max}$ (7)	from $0.35E^{\max}/4$ to P^{abs}
added	ψ	harm per unit of excess protein share (to source); enters the health burden

What changes in results: the fishing village is no longer stopped at an arbitrary protein amount. It eats up to the liver's capacity and carries the harm of a 50 % protein diet.

Proposal F. Energy balance with a body store (fixes F12)

Reasoning:

- Energy eaten is either spent or stored.
- People fatten in good seasons and live off reserves in lean ones. That is the buffer that keeps a bad harvest from becoming an immediate famine, and it is absent now.
- With a store, surplus energy is not all turned into activity. Activity follows work; surplus is stored up to a ceiling; deficits are drawn from the store at a limited rate.

Equations:

$$\Lambda^{\text{act}} = \min\!\Big(\Lambda^{\max},\ \frac{E + \min(Z/365,\ \bar{z})}{B}\Big) \ \text{capped at the work demand } \Lambda^{\text{w}} \ \text{(plus the voluntary surplus, M85)}$$
$$Z_{t+1} = \min\!\big(Z^{\max},\ Z_t + 365\,(E - B\Lambda^{\text{act}})\big)$$

Variables:

Change	Variable	Note
added	Z	body energy store per person (a state carried from year to year, like the vitamin stores of `round_m85.md`)
added	$Z^{\max}$, $\bar{z}$	store ceiling and the largest daily drawdown (to source)
changed	Λ^{act} (25)	from E/B to the balance above

Proposal G. Work by kind of work, and what caffeine does (fixes F13, F14)

Reasoning:

- Physical work scales with the energy available for it.
- Desk work is lowered by a deficit but not raised by a surplus.
- Caffeine raises effort and alertness (PMID 20164571; Dulloo 1989 for the energy cost), so it belongs in work capacity as well as in the need.

Equations:

$$work_k = \begin{cases} \min\big(\Omega + \omega_c c,\ \max(0,\ \frac{\Lambda^{\text{act}} - \Lambda^{\min}}{\Lambda_k - \Lambda^{\min}})\big) & k \in \{\text{vigorous, active}\} \\ \min\big(1 + \omega_c' c,\ x\big) & k = \text{sedentary} \end{cases}$$

Variables:

Change	Variable	Note
changed	$work$ (26)	becomes $work_k$ per kind of work
added	$\omega_c,\ \omega_c'$	caffeine's effect on physical and mental work capacity (to source)

Proposal H. Water as a balance in consistent units (fixes F15, F16)

Reasoning:

- A drink contributes the water it contains, times how well that water is retained. The hydration index is that retention, relative to water.
- The need rises with energy turnover (work, heat). Its slope still has to be sourced; I found no confirmed figure in this round.

Equations:

$$W^{\text{drink}} = \sum_{g \in \mathcal{G}^{\text{D}}} q_g \eta_g \frac{\omega_g}{100} h_g, \qquad W^{\text{need}} = W_0 + w_E \left(E^{\text{need}} - E_0\right)$$

For milk: $0.88 \times 1.5 = 1.32$ instead of 1.5.

Variables:

Change	Variable	Note
changed	W^{drink} (23)	water content × retention
changed	W^{need}	from a constant to a function of energy turnover
added	W_0, E_0, w_E	base need at a reference energy, and the slope (to source)

Proposal I. Seasons for fresh goods (fixes F17)

Reasoning:

- Fresh goods are available in their season only.
- With the vitamin stores of `round_m85.md`, a city that eats fruit only in summer can run short of vitamin C by late winter, and the model should be able to show it.
- Two half-years are the smallest split that shows it.

Equations:

$$a_{g,\tau} = \frac{a_g}{|\mathcal{T}_g|/|\mathcal{T}|} \text{ for } \tau \in \mathcal{T}_g,\ \ 0 \text{ otherwise;} \qquad \text{the program runs once per } \tau \in \mathcal{T} = \{\text{summer}, \text{winter}\}$$

Variables:

Change	Variable	Note
added	$\mathcal{T}$, index τ	seasons
added	$\mathcal{T}_g$	the seasons in which good g is fresh

Change	Variable	Note
changed	a_g	becomes $a_{g,\tau}$; nutrient stores carry between seasons

Proposal J. Classes within the city (fixes F18), then prices (F19)

Reasoning:

- Food is not shared equally. Three classes are the least that show a fed average hiding a starving poor.
- When the goods economy has prices, each class's access becomes its budget, and the classic diet problem (cheapest diet meeting the needs, Stigler 1945) becomes a constraint of the same program.

Equations:

$$\text{for each class } j: \quad a_g^{(j)} = a_g \, \frac{\theta_j^{\text{access}}}{\theta_j^{\text{pop}}}; \qquad \text{later: } \sum_g p_g q_g^{(j)} \le Y_j$$

Variables:

Change	Variable	Note
added	class index $j \in \{\text{poor}, \text{middle}, \text{rich}\}$	
added	θ_j^{pop}, θ_j^{access}	population share and share of food access (from income distribution, to source)
added later	p_g, Y_j	prices and budgets from the goods economy
changed	all outcomes	computed per class; the population model reads them per class or weighted

3. The model after A-E, in one statement

Given: $N,\ \mathbf{S},\ \mathbf{w};$ parameters of section 3 of the specification, $\omega_i,\ b_{gi},\ h_g.$

Stage 1: $A^* = \max_{\mathbf{q}} \sum_{i \in \mathcal{I}_0} \omega_i \min\big(1, c_i(\mathbf{q})\big)$

Stage 2: $\max_{\mathbf{q}} \sum_g h_g \ln(1 + q_g/m_g) - \zeta \max(0, \text{alcohol} - \alpha^{\text{safe}}) \quad \text{s.t. } A(\mathbf{q}) \ge A^*$

subject to: $0 \le q_g \le \bar{q}_g,\ \sum_g \sigma_g e_g q_g \le F^{\max},\ \sum_g e_g q_g \le E^{\max},\ \sum_g \hat{\pi}_g q_g \le P^{\text{abs}},\ \sum_{g \in \mathcal{G}^{\text{F}}} \eta_g q_g \le Q^{\text{fruit}}$

Outcomes: $x = E/E^{\text{need}},\ fed = F/F^{\text{fed}},\ c_i,\ \Psi,$ protein harm, $W^{\text{plain}},\ work_k,\ L_g,\ D_g$ $(+Z$ with F$)$

Compared with version 3:

- one satiety construct instead of three;
- weights that come from health (ω_i) instead of a 1 : 1 sum;
- diminishing returns where they occur (in nutrient gaps and in repeated pleasures);
- an outcome for well-being;
- two priorities stated as what they are: a lexicographic order.

4. Order of decision

Order	Proposal	Why first
1	A, C, D	they define what “nutrition” and “priority” mean; the rest builds on them
2	B	well-being per serving and its outcome
3	E, H	corrections of units and ceilings
4	G, F	work and the body store
5	I, J	seasons and classes add dimensions; they cost the most data and computation